MID-TERM EXAMINATION
(Course Name B. Tech. CSE/IT/CSEAI) (Semester 4th)
(February, 2023) OFF LINE mode

Subject Code: BIT 202	Subject: Operating Systems
Time: 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1 (2.5*4)

(a) Define system call and various types of system calls provided by operating system with examples.

(b) Explain the concept of 'process'. Also, describe the contents of a process control block (PCB).

(c) Differentiate between physical and logical memory address space with example.

(d) Define Page fault. Explain Page replacement using demand paging with block diagram.

Q2 (Attempt any Two Parts) (5,5)

(a) For the following set of processes, find the average waiting time using Gantt chart for:

(i) SJF

(ii) Priority scheduling process:

	Burst time	Priority
P_1	5	5
P_2	3	4
P_3	8	3
P_4	2	1
P_5	1	2

The processes have arrived in the order P_2, P_1, P_4, P_3 and P_5

(b) Find the turn-around time and average waiting time for the following processes for Round Robin Scheduling:

Given : Time quantum = 2 secs

Process id	Arrival time	Burst time
P_1	0	5
P_2	1	3
P_3	2	1
P_4	3	2
P_5	4	3

(c) For the following set of processes, find the average waiting time & average turn-around time using GANTT chart for:

SJF preemptive algorithm

Process	Arrival time (in sec)	Burst Time (in sec)
P_1	0	4
P_2	1	2
P_3	2	5
P_4	3	4

Q3 (Attempt any Two Parts) (5,5)

(a) Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. the number of frames in the memory is 3. Find out the number of page faults respective to:

1. Optimal Page Replacement Algorithm
2. LRU Page Replacement Algorithm

(b) Consider the following page reference string:

3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3

Find the number of page faults with FIFO and LRU, where 5 page frames are present.

(c) What is Deadlock? What are the characteristics of deadlock? Explain different methods for handling it in detail.